15 g is a function such that

$$g(x) = x^3 + 1$$

(a) Find g(–3)

$g(-3) = (-3)^3 + 1 = -27 + 1 = -26$

1 mark
Substitute −3.

.....

(1)

(b) Find $g^{-1}(x)$

$y = x^3 + 1 \rightarrow y - 1 = x^3$

$g^{-1}(x) = \sqrt[3]{(x - 1)}$

1 mark
Subtract 1.

1 mark
Then cube root.

.....

(2)

(Total for Question 15 is 3 marks)